

Eenheid 1.7

Note Title

2015/02/09

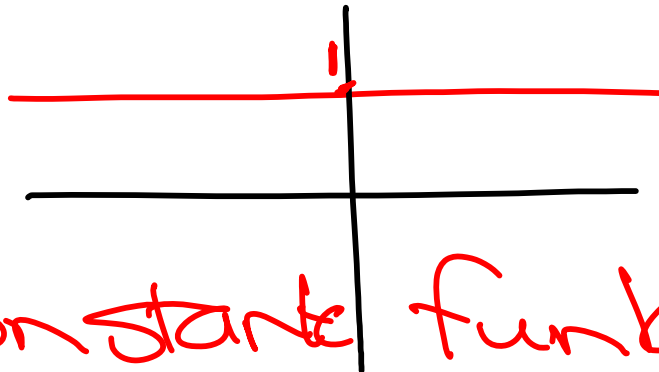
Exponentiaal Functie pp 45-55

Definisie: 'n Eksponentiaal-funksie is 'n funksie in die vorm

$$f(x) = a^x, a > 0$$

3 Soorte:

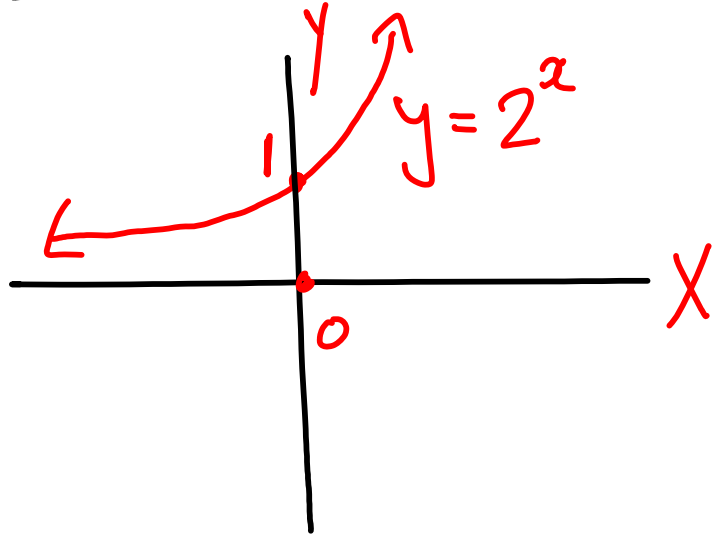
① As $a = 1$



$$y = 1^x$$

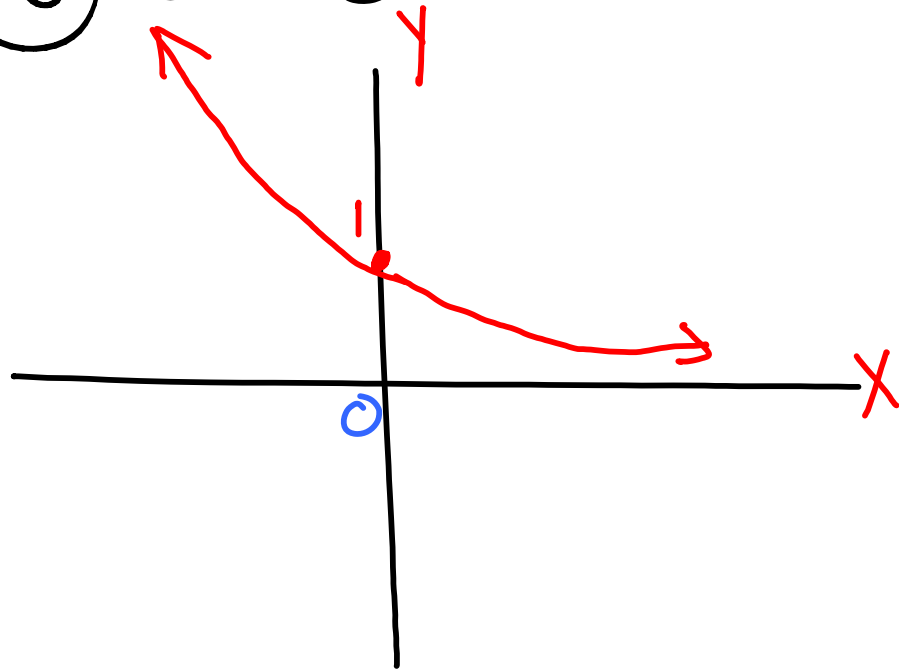
f is dan 'n konstante funksie

② as $a > 1$ But $a^x = 2^x$



Eksponsiale toename

③ as $0 < a < 1$



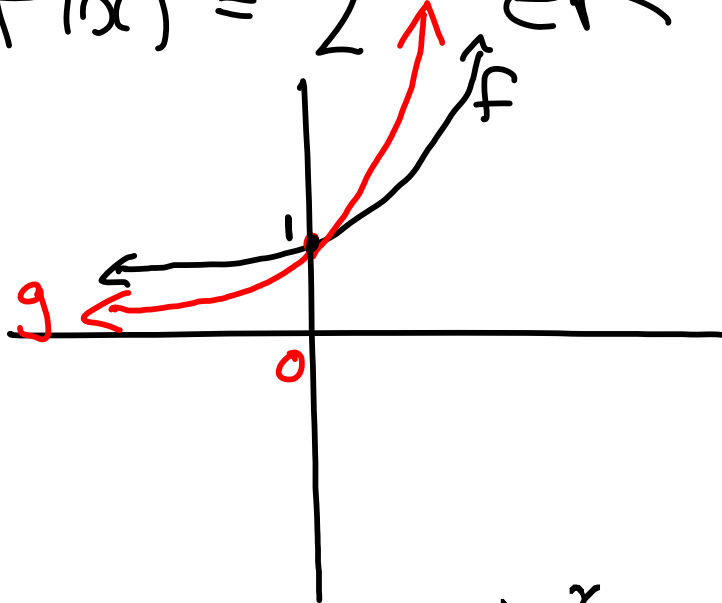
$$y = \left(\frac{1}{2}\right)^x = \left(\underline{2}^{-1}\right)^x$$
$$y = \underline{2}^{-x}$$

Eksponsiale afname

Op die selfde assenstelsel

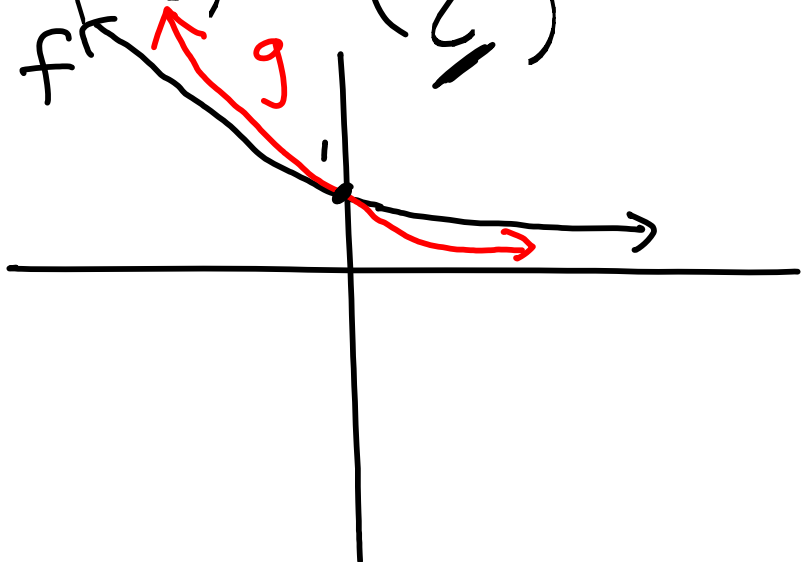
• Skets

① $f(x) = 2^x$ en $g(x) = 3^x$

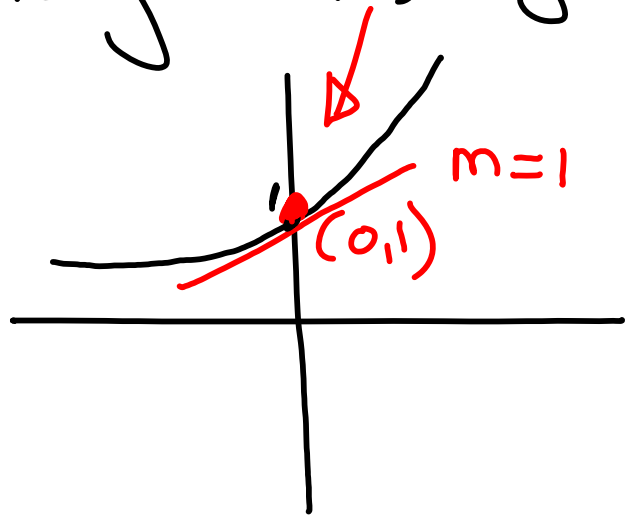


② $f(x) = \left(\frac{1}{2}\right)^x$ en $g(x) = \left(\frac{1}{3}\right)^x = 3^{-x}$

$f(x) = \left(\frac{1}{2}\right)^x = 2^{-x}$



in Switserse Wiskundige Euler (1727)
 het gesoek na 'n getal a waar die
 helling van die raaklyn aan $y=a^x$
 by $(0,1)$ gelyk is aan 1.

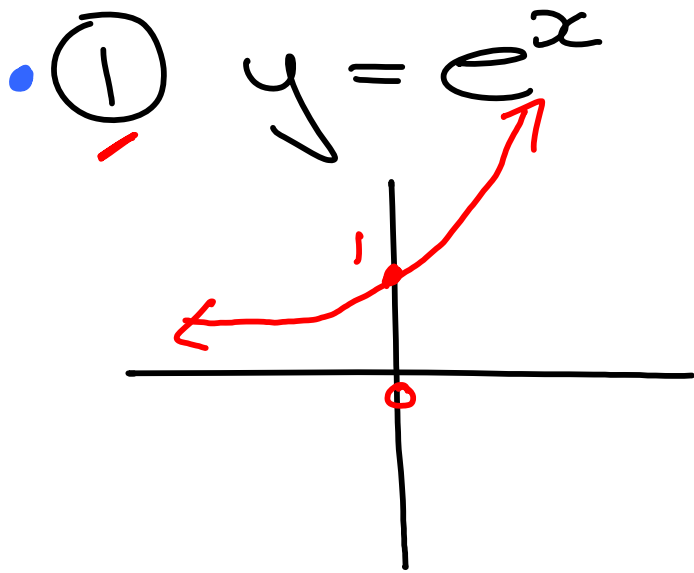


Euler het die getal
 gekry. e (1727)
 3

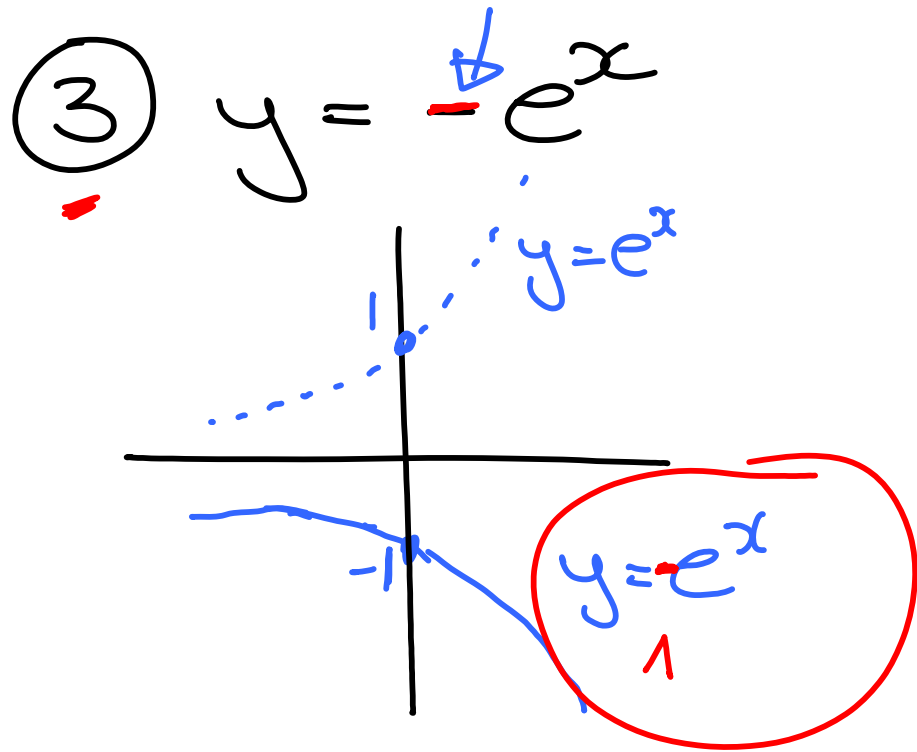
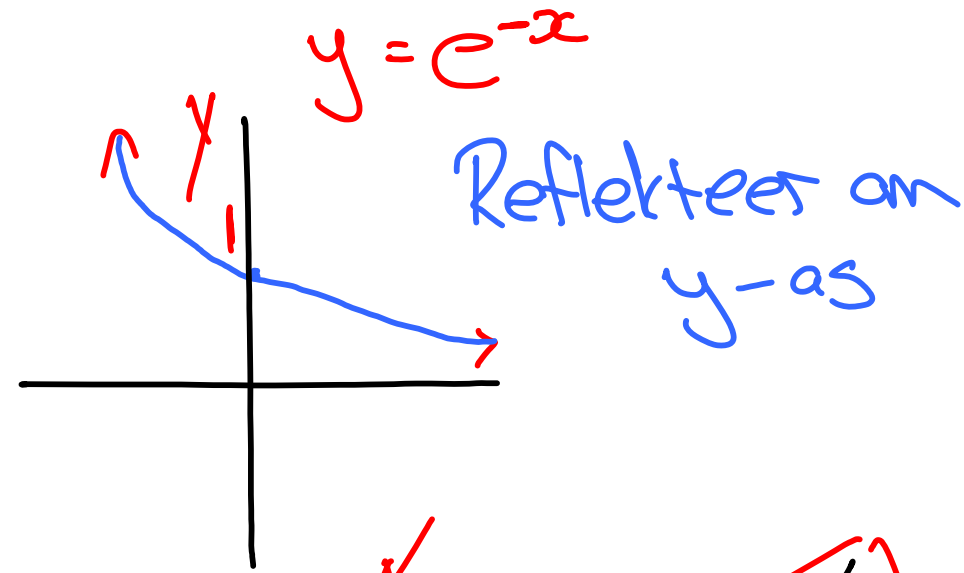
Sakrekenaar: $e \approx 2.71828\dots$

$$2 < e < 3$$

$$\Rightarrow 2^x < e^x < 3^x$$

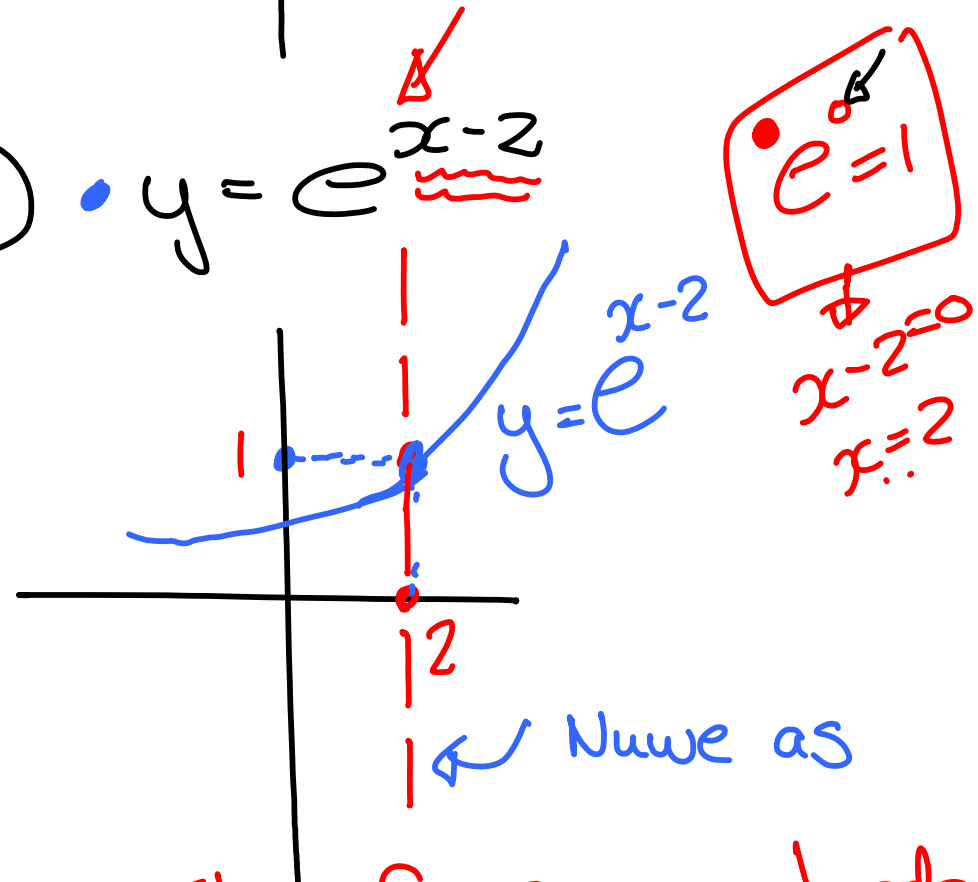


②



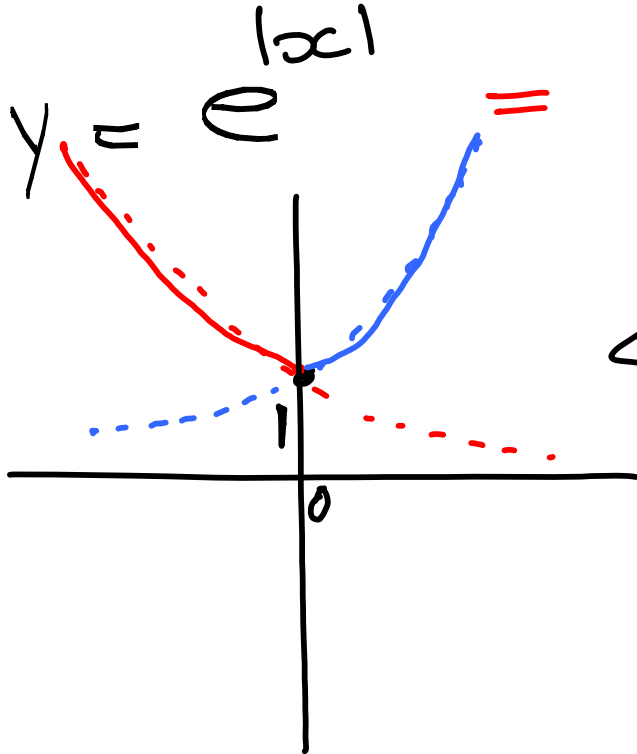
Reflekteer om x-as

④

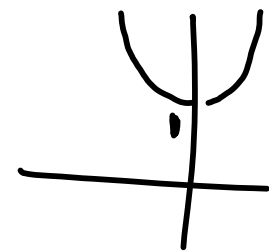


• Skuif 2 eenhede regs.

5

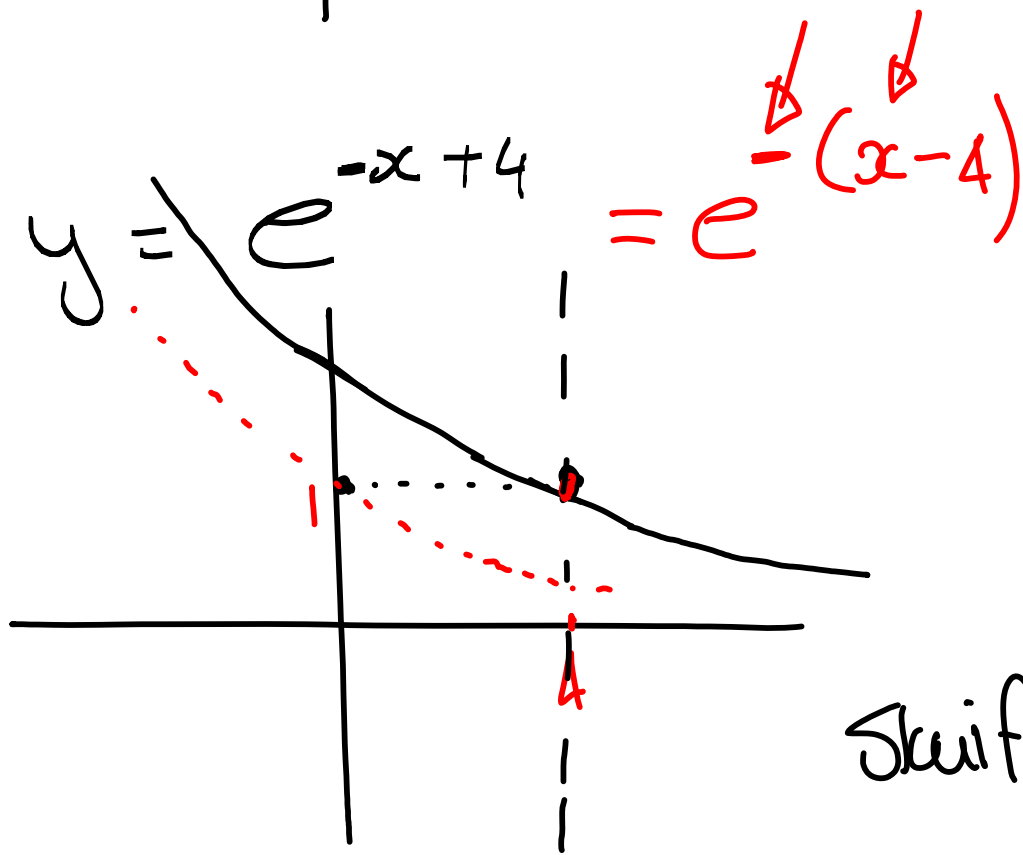


$$\begin{cases} e^x & \text{as } x \geq 0 \\ e^{-x} & \text{as } x < 0 \end{cases}$$



Skets : gebruik def.
van $|x|$

6



- Nuwe as $x=4$
- Inpunt $e^0 = 1$

$$-x+4=0$$

$$-x = -4$$

$$x = 4$$

Skuif $y = e^{-x}$ 4 eenhede na regs

Exponentwetste

p 47

- a, b positiewe getalle
- $x, y \in \mathbb{R}$

$$1. \quad \underline{a}^{x+y} = a^x a^y$$

$$2. \quad \underline{a}^{x-y} = \frac{a^x}{a^y}$$

$$3. \quad (\underline{a}^x)^y = a^{xy}$$

$$4. \quad (ab)^x = a^x b^x$$

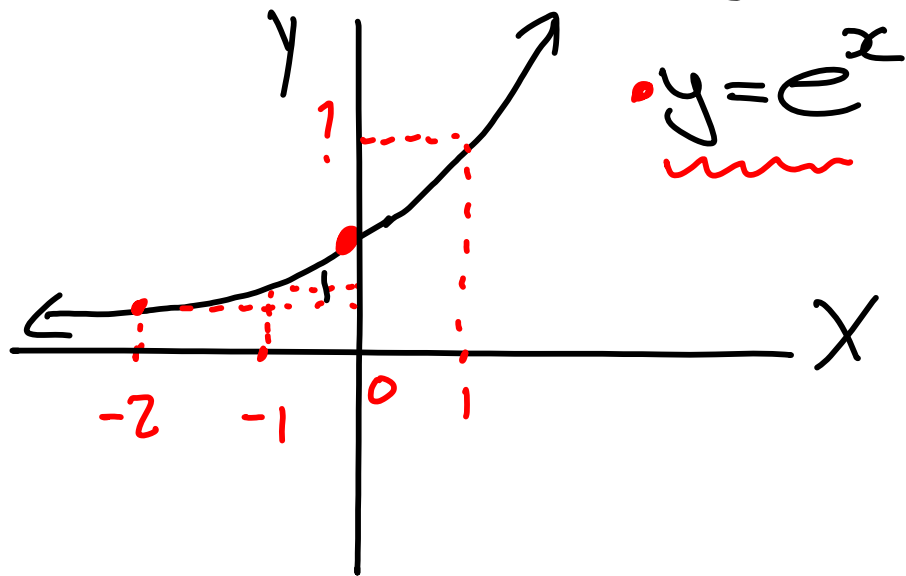
Voorbeelde

$$1. \quad \frac{3^x 3^{6x}}{3^{4x}} = \frac{3^{7x}}{3^{4x}} = 3^{3x}$$

$$2. \quad \frac{e^3 \sqrt{e^4}}{e^8} = \frac{e^3 e^2}{e^8} = \frac{e^5}{e^8} = \boxed{\frac{1}{e^3} = e^{-3}}$$

$$3. \quad \frac{(e^x e^y)^4}{e^{5x} e^{9y}} = \frac{e^{4x} e^{4y}}{e^{5x} e^{9y}} = e^{4x-5x} e^{4y-9y} = e^{-x-5y}$$

Opmerking:



- $W_f = (0, \infty) = \{y \mid y > 0\}$
- $D_f = \mathbb{R}$
- $e^0 = 1$
- $e^1 = 2.718$
- $e^{-1} = \frac{1}{e} = \frac{1}{2.718}$
- $e^{-2} = \frac{1}{e^2} = \frac{1}{(2.718)^2}$